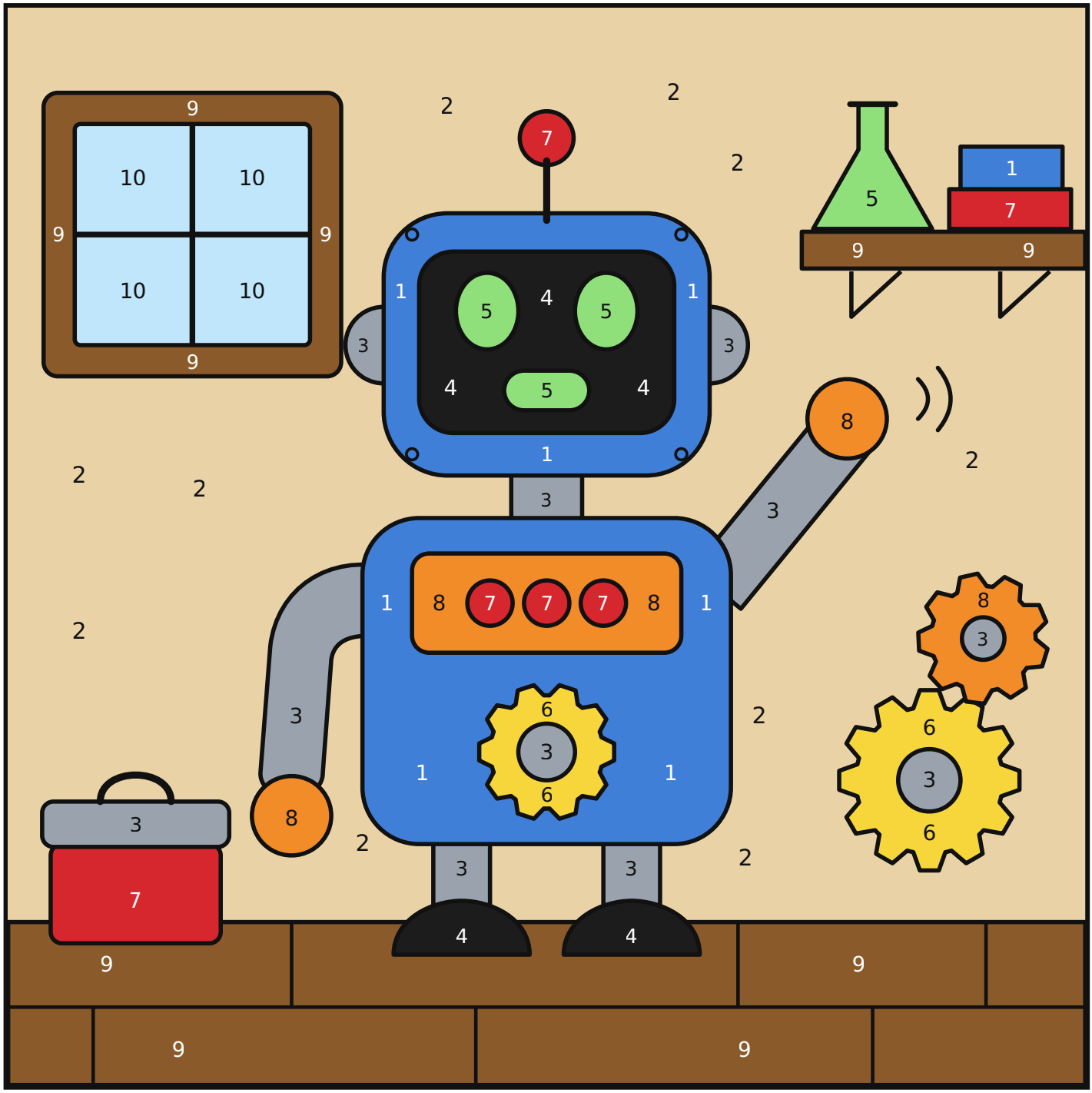
Name: **ANSWER KEY** _____ Date: _____

Basics & Reasoning: Color By Number



Color Code:

8 Orange 11	**6** Yellow (3, 2)	**10** Light Blue 15	**4** Black 82°	**3** Gray 76°
7 Red 77°	**5** Light Green 5	**9** Brown 106°	**1** Blue 28	**2** Tan 65°

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Basics & Reasoning: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
M is the midpoint of AB. AM = 3x + 2 and MB = 5x − 6. Find the length of AB.

3x + 2 = 5x − 6, so x = 4
AM = 14, so AB = 2(14)
28

1. Color: **Blue**

2.
Two angles are complementary. One measures x° and the other measures (2x + 15)°. Find the larger angle.

x + 2x + 15 = 90, so x = 25
larger = 2(25) + 15
65°

2. Color: **Tan**

3.
Two angles are supplementary and measure (3x + 10)° and (5x − 6)°. Find the smaller angle.

8x + 4 = 180, so x = 22
angles: 76° and 104°
76°

3. Color: **Gray**

4.
Ray BD bisects ∠ABC. m∠ABD = (3x + 5)° and m∠DBC = (5x − 19)°. Find m∠ABC.

3x + 5 = 5x − 19, so x = 12
∠ABD = 41°, so ∠ABC = 2(41)
82°

4. Color: **Black**

5.
Find the distance between (1, 2) and (4, 6).

$d = \sqrt{(3^2 + 4^2)} = \sqrt{25}$
5

5. Color: **Light Green**

6.
Find the midpoint of the segment with endpoints (−2, 6) and (8, −2).

x: $\frac{-2 + 8}{2} = 3$, y: $\frac{6 + (-2)}{2} = 2$
(3, 2)

6. Color: **Yellow**

7.
Two vertical angles measure (4x − 7)° and (3x + 14)°. Find the measure of each angle.

4x − 7 = 3x + 14, so x = 21
4(21) − 7
77°

7. Color: **Red**

8.
Points A, B, C are collinear with B between A and C. AB = 12, BC = 2x + 1 and AC = 5x − 2. Find BC.

12 + 2x + 1 = 5x − 2, so x = 5
BC = 2(5) + 1
11

8. Color: **Orange**

9.
Two angles form a linear pair and measure (2x + 20)° and (4x − 2)°. Find the larger angle.

6x + 18 = 180, so x = 27
angles: 74° and 106°
106°

9. Color: **Brown**

10.
How many different line segments can be named using any two of 6 collinear points as endpoints?

each pair of points makes one segment: $\frac{6 \cdot 5}{2}$
15

10. Color: **Light Blue**